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MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE

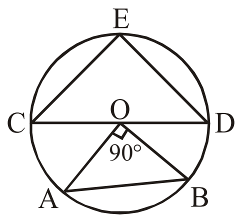
Exercise

Fundamental of Geometry

(English Medium)

EXERCISE

1. In the diagram 'O' is the centre of the circle. The point E lies on the circumference of the circle such that the area of the ΔECD is maximum. If $\angle AOB$ is a right angle, then the ratio of the area of ΔECD to the area of the ΔAOB is :



(A) 2 : 1 (B) 3 : 2 (C) 4 : 3 (D) 4 : 1

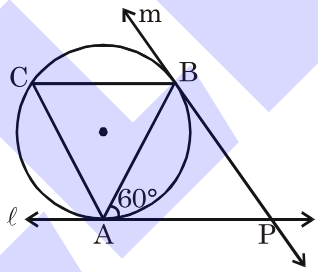
GM0001

2. Let ΔXOY be a right angled triangle with $\angle XOY = 90°$. Let M and N be the midpoints of legs OX and OY, respectively. Given that XN = 19 and YM = 22, the length XY is equal to

(A) 24 (B) 26 (C) 28 (D) 34

GM0002

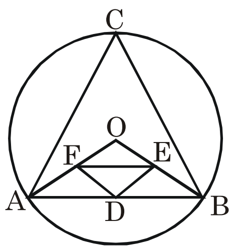
3. In the diagram below, if ℓ and m are two tangents and AB is a chord making an angle of 60° with the tangent ℓ, then the angle between ℓ and m is



(A) 45° (B) 30° (C) 60° (D) 90°

GM0003

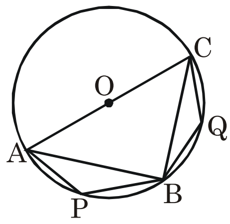
4. In the diagram, O is the centre of the circle and D, E and F are mid points of AB, BO and OA respectively. If $\angle DEF = 30°$, then $\angle ACB$ is



(A) 30° (B) 60° (C) 90° (D) 120°

GM0004

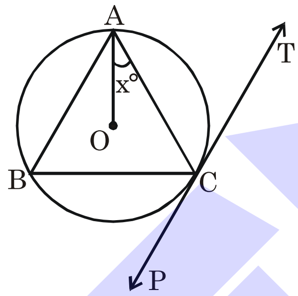
5. In the below diagram, O is the centre of the circle, AC is the diameter and if $\angle APB = 120°$, then $\angle BQC$ is



(A) 30° (B) 150° (C) 90° (D) 120°

GM0005

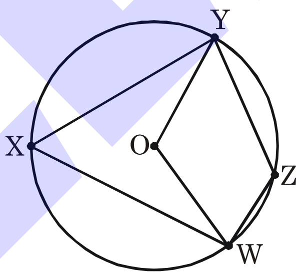
6. In the adjoining figure, PT is a tangent at point C of the circle. O is the circumcentre of ΔABC. If $\angle ACP = 118°$, then the measure of $\angle x$ is



(A) 28° (B) 32° (C) 42° (D) 38°

GM0006

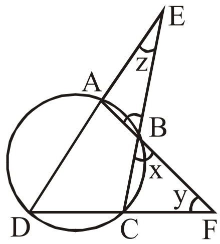
7. In the cyclic quadrilateral WXYZ on the circle centered at O, $\angle ZYW = 10°$ and $\angle YOW = 100°$. What is the measure of $\angle YWZ$?



(A) 30° (B) 40° (C) 50° (D) 60°

GM0007

8. In the adjacent figure, if $\angle y + \angle z = 100°$ then the measure of $\angle x$ is :



(A) 50° (B) 40° (C) 45° (D) Cannot be determined

GM0008

9. In a ΔABC if P be the point of intersection of the interior angle bisectors and Q be the point of intersection of the exterior angle bisectors of angles B and C respectively, then the figure BPCQ is a :

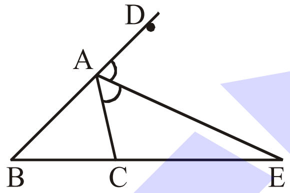
(A) Parallelogram (B) Rhombus (C) Trapezium (D) Cyclic quadrilateral

GM0009

10. D, E and F are points on sides BC, CA and AB respectively of ΔABC such that AD bisects $\angle A$, BE bisects $\angle B$ and CF bisects $\angle C$. If AB = 5 cm, BC = 8 cm and CA = 4 cm, determine AF, CE and BD.

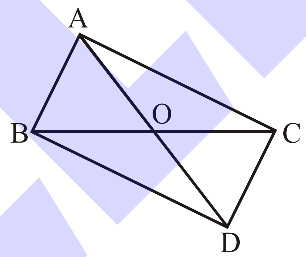
GM0010

11. In the adjoining figure, AE is the bisector of exterior $\angle CAD$ meeting BC produced in E. If AB = 10 cm, AC = 6 cm and BC = 12 cm, find CE.



GM0011

12. In fig. ΔABC and ΔDBC are two triangles on the same base BC. Prove that $\dfrac{ar(\Delta ABC)}{ar(\Delta DBC)} = \dfrac{AO}{DO}$.

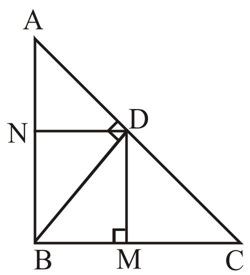


GM0012

13. ABC is a right-triangle with $\angle ABC = 90°$, $BD \perp AC$, $DM \perp BC$ and $DN \perp AB$. Prove that

(i) $DM^2 = DN \times MC$

(ii) $DN^2 = DM \times AN$

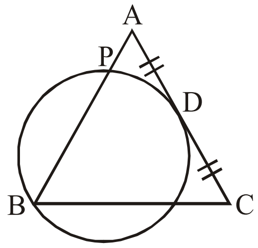


GM0013

14. Prove that sum of squares of diagonals of a parallelogram is equal to sum of squares of its sides.

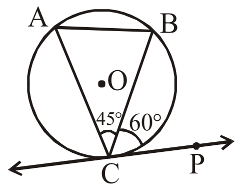
GM0014

15. In fig., ABC is a triangle in which AB = AC. A circle through B touches AC at D and intersects AB at P. If D is the mid-point of AC, show that 4 AP = AB.



GM0015

16. In fig., CP is a tangent to a circle. If $\angle PCB = 60°$ and $\angle BCA = 45°$, find $\angle ABC$.



GM0016

17. The inscribed circle of ΔABC touches BC, CA and AB at X, Y and Z respectively. If $\angle A = 64°$, $\angle C = 52°$, find $\angle XYZ$ and $\angle XZY$.

GM0017

18. Let $\overline{AM}$ be a median of ΔABC. Prove that $AM > (AB + AC - BC)/2$.

GM0018

ANSWER KEY

Do yourself-1

1. 1 **3.** 5 : 4 **4.** (i) $\frac{5}{8}$ (ii) $\frac{25}{64}$ 5. C 6. 5 : 9

Do yourself-2 :

2. 2cm **3.** $\sqrt{5}$ **4.** 30º 5. B 6. B

Do yourself-3 :

1. 35º **2.** 3cm **3.** 9.6 cm **4.** 168 5. C 6. B

Do yourself-4 :

1. 72 **2.** x = 2 cm **3.** 90º 4. C 5. 168

EXERCISE

1.	A	**2.**	B	**3.**	C	**4.**	B	**5.**	B	**6.**	A
7.	B	**8.**	B	**9.**	D	**10.**	AF = $\frac{5}{3}$, CE = $\frac{32}{13}$, BD = $\frac{40}{9}$				

11. CE = 18 cm **16.** $\angle ABC = 75°$ **17.** $\angle XYZ = 58°$, $\angle XZY = 64°$